



OPEN Benchmarking MedMNIST dataset on real quantum hardware

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Quantum machine learning (QML) has emerged as a promising domain to leverage the computational capabilities of quantum systems to solve complex classification tasks. In this work, we present the first comprehensive QML study by benchmarking the MedMNIST—a diverse collection of medical imaging datasets on a 127-qubit real IBM quantum hardware, to evaluate the feasibility and performance of quantum models (without any classical neural networks) in practical applications. This study explores recent advancements in quantum computing such as device-aware quantum circuits, error suppression, and mitigation for medical image classification. The proposed methodology is comprised of three stages: preprocessing, generation of noise-resilient and hardware-efficient quantum circuits, optimizing/training of quantum circuits on classical hardware, and inference on real IBM quantum hardware. Firstly, we process all input images in the preprocessing stage to reduce the spatial dimension due to quantum hardware limitations. We generate hardware-efficient quantum circuits using backend properties expressible to learn complex patterns for medical image classification. After classical optimization of QML models, we perform inference on real quantum hardware. We also incorporate advanced error suppression and mitigation techniques in our QML workflow, including dynamical decoupling (DD), gate twirling (Twir), and matrix-free measurement mitigation (M3) to mitigate the effects of noise and improve classification performance. The experimental results showcase the potential of quantum computing for medical imaging and establish a benchmark for future advancements in QML applied to healthcare.

Biomedical image analysis plays a critical role in advancing human healthcare. Recently, machine learning (ML) has demonstrated remarkable potential in this domain, excelling in tasks such as medical image classification^{1–3}, segmentation^{4,5}, and reconstruction^{6,7}. Moreover, QML, an emerging paradigm of ML implemented on quantum computers, is anticipated to offer quantum advantages over classical ML. These advantages include enhanced speed-ups, improved model expressivity, efficient exploration of high-dimensional feature spaces, and the ability to address certain classically intractable problems^{8–12}. QML capitalizes on the unique properties of quantum mechanics, such as superposition, entanglement, and interference, to unravel complex relationships between inputs and labels. In the current noisy intermediate-scale quantum (NISQ) era, QML faces notable challenges. Limited access to large-scale quantum systems constrains the ability to encode high-dimensional data into quantum states. Hardware noise and decoherence further impact accuracy and restrict circuit depth, thereby limiting the expressivity of QML models. Rapid advancements in quantum hardware, including improved quantum error correction techniques¹³, are expected to overcome these limitations and enable the practical deployment of QML for real-world applications in the near future.

Despite these obstacles, QML research has witnessed significant growth^{14–28}. Generally, QML approaches are based on parameterized quantum circuits (PQCs)²⁹ consisting of fixed gates (such as SWAP and CNOT) and single-qubit (such as Pauli rotation) gates with trainable parameters. These PQCs are comprised of three stages: data encoding, a variational ansatz circuit with trainable parameters, and measurement to classical bits. The data encoding stage encodes the classical input data to quantum states. The parameters of the ansatz circuit are optimized using classical gradient descent algorithms, typically relying on the parameter-shift rule³⁰, enabling the circuit to solve machine learning tasks.

Over the last few years, researchers have explored QML algorithms for medical imaging in various studies including^{26,31–39} to harness potential quantum advantages over classical machine learning approaches. These studies address a variety of medical imaging modalities, including MRI, CT, X-ray, OCT, and ultrasound, targeting tasks such as clinical data classification and detection. The QML algorithms investigated in these works include quantum support vector machines (SVM)³⁴, hybrid quantum-classical neural networks^{26,35,36,38,39}, quantum transfer learning^{33,37}, and variational quantum circuits^{31,32}. To the best of our knowledge, few studies such as^{26,33,34,37,39} have used real quantum hardware for evaluation purposes but on a limited scale. Cherrat et al. proposed a quantum vision transformer (QViT) in²⁶ based on shallow PQCs capable of performing complex

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image classification tasks. QViT introduces trainable quantum orthogonal layers adaptable to varying levels of quantum hardware connectivity. The authors have evaluated QViT on open-source MedMNIST (a collection of 12 biomedical imaging datasets)¹ using a simulator and real hardware results are provided for only one dataset (RetinaMNIST) out of 12 total. In³³, Acar and Yilmaz designed a hybrid classical-quantum transfer learning based QML model for COVID-19 detection using CT images. The authors claimed to achieve a classification accuracy of 94–100% on quantum devices with limited qubit capacity compared to 90% on classical systems using a small dataset. This highlights the potential of quantum computers for efficient and accurate medical image classification, particularly with limited data. Moradi et al.³⁴ designed two QML methods i.e., quantum distance classifier and quantum-kernel SVM for clinical data classification on quantum hardware. The findings observed in this study also highlight the promise of QML in clinical data analysis, especially for tasks constrained by dataset size and feature complexity. Azevedo et al.³⁷ employed a hybrid quantum-classical neural network for mammogram classification into malignant and benign categories. The authors compared the results obtained from the quantum device and simulator, highlighting the potential of QML in medical diagnostics while emphasizing the need for further validation. Similarly, the study conducted in³⁹ by Landman et al. introduces two quantum neural networks: quantum-assisted neural networks, which leverage the quantum computer to estimate inner products during the training and inference of classical neural networks, and quantum orthogonal neural networks, built on quantum pyramidal circuits for orthogonal matrix multiplication. This study also considered only two datasets (PneumoniaMNIST and RetinaMNIST) out of 12 datasets from MedMNIST for evaluation purposes on real hardware. Note that most of these QML studies are based on hybrid quantum-classical neural networks that combine quantum circuits with classical neural networks, leaving the potential of purely quantum approaches underexplored. Also, the individual contribution of both quantum and classical parts is unclear. While some studies claim potential quantum advantages, these are often observed only in restricted settings (e.g., small datasets) with limited evidence of their generalization across different image classification domains or modalities. Demonstrating clear, reproducible quantum advantages over classical machine learning models remains a significant challenge.

To address above mentioned limitations, we conducted a comprehensive QML study for medical image classification on a 127 qubits IBM quantum hardware. This research study aims to explore the different aspects of QML for image classification including efficient encoding of high-dimensional image data to quantum states, generation of device-aware quantum circuits having optimal expressivity for different image datasets, advantages of error suppression and mitigation techniques. By exclusively employing quantum methods without classical neural networks, our work addresses critical gaps in the current QML literature, offering insights into the practical applicability and performance of quantum-only models for medical image classification tasks. This effort not only highlights the promise of QML in addressing real-world problems but also sets a foundational benchmark for future explorations in this domain. The main contributions of our research study include: (1) We present the first comprehensive QML study by benchmarking the MedMNIST dataset on real IBM Cleveland hardware, (2) This research leverages and evaluates the recent quantum computing capabilities such as generation of device-aware quantum circuits and error suppression/mitigation techniques for medical image classification, (3) We also conducted an ablation study on different error suppression and mitigation techniques to provide an intuitive understanding, and (4) The experimental results provide foundational insights into the performance and scalability of quantum models for medical image classification, establishing a benchmark for future QML research for healthcare applications.

Results

Dataset and evaluation metrics details

In this subsection, we provide the details of the MedMNIST datasets¹ used in this benchmark study and performance metrics used to evaluate the QML workflow. The MedMNIST contains 12 two-dimensional biomedical imaging datasets of different medical modalities and complexity levels. These 12 datasets include 2-class as well as multi-class datasets including 4-class, 5-class, 7-class, 8-class, 9-class, and 11-class. For this study, we considered 8 datasets out of the 12 by making sure that at least one dataset from each class category was considered as shown in Table 1. The different medical modalities considered in these 8 datasets are illustrated in Fig. 1. We considered two widely used classification metrics i.e., accuracy (ACC) and area under the curve (AUC) to measure the performance of QML models. It is important to note that ACC provides a straightforward interpretation of model performance but it is highly sensitive to class imbalance, often leading to biased evaluations. On the contrary, AUC offers a more comprehensive evaluation of the classifier's separability and is inherently robust to class imbalance. This makes AUC particularly suitable for datasets where class distributions are skewed. It is clear from Table 1 that the MedMNIST datasets are quite imbalanced. Therefore, we prioritized AUC over ACC to select a model for real hardware execution. By focusing on AUC, we ensured that the chosen quantum models were able to effectively distinguish between classes across various thresholds, providing a fairer and more reliable assessment of performance. The classical post-processing of M3-mitigated bitstrings in the inference stage provides an expectation values vector for all test samples. These expectation values are used to calculate the ACC and AUC metrics. For binary datasets (where we measure one qubit), the accuracy is calculated by measuring the proportion of predictions that deviate from the true labels by less than a threshold of 1. It evaluates how many predictions are close enough to the actual labels, providing an overall measure of correctness for the model's outputs in binary classification tasks. The binary AUC is computed by assessing the separability of the two classes on the basis of the predicted expectation values. We generate signed prediction labels to align with the true labels and calculate the AUC score using the AUC score function⁴⁰ from scikit-learn. For the multi-class datasets (where we are measuring more than 2 qubits), the model's predicted class is determined by taking the index of the maximum value along each row of the expectation values vector. This is compared against the true labels to compute the classification accuracy using

MedMNIST2D	Data modality	Classes	Training/test	Sample distribution
PneumoniaMNIST	Chest X-Ray	2	4708/624	Train: 1214, 3494 Test: 234, 390
BreastMNIST	Breast Ultrasound	2	546/156	Train: 147, 399 Test: 42, 114
OCTMNIST	Retinal OCT	4	97,477/1000	Train: 33484, 10213, 7754, 46026 Test: 250, 250, 250, 250
RetinaMNIST	Fundus Camera	5	1080/400	Train: 486, 128, 206, 194, 66 Test: 174, 46, 92, 68, 20
DermaMNIST	Dermatoscope	7	7007/2005	Train: 228, 359, 769, 80, 779, 4693, 99 Test: 66, 103, 220, 23, 223, 1341, 29
BloodMNIST	Blood Cell Microscope	8	11,959 / 3421	Train: 852, 2181, 1085, 2026, 849, 993, 2330, 1643 Test: 244, 624, 311, 579, 243, 284, 666, 470
PathMNIST	Colon Pathology	9	89,996 / 7180	Train: 9366, 9509, 10360, 10401, 8006, 12182, 7886, 9401, 12885 Test: 1338, 847, 339, 634, 1035, 592, 741, 421, 1233
OrganSMNIST	Abdominal CT	11	13,932 / 8827	Train: 1148, 630, 614, 721, 1132, 1119, 3464, 741, 803, 2004, 1556 Test: 811, 439, 445, 510, 704, 693, 2078, 397, 439, 1343, 968

Table 1. Data summary of MedMNIST-v2 datasets used in the QML workflow.

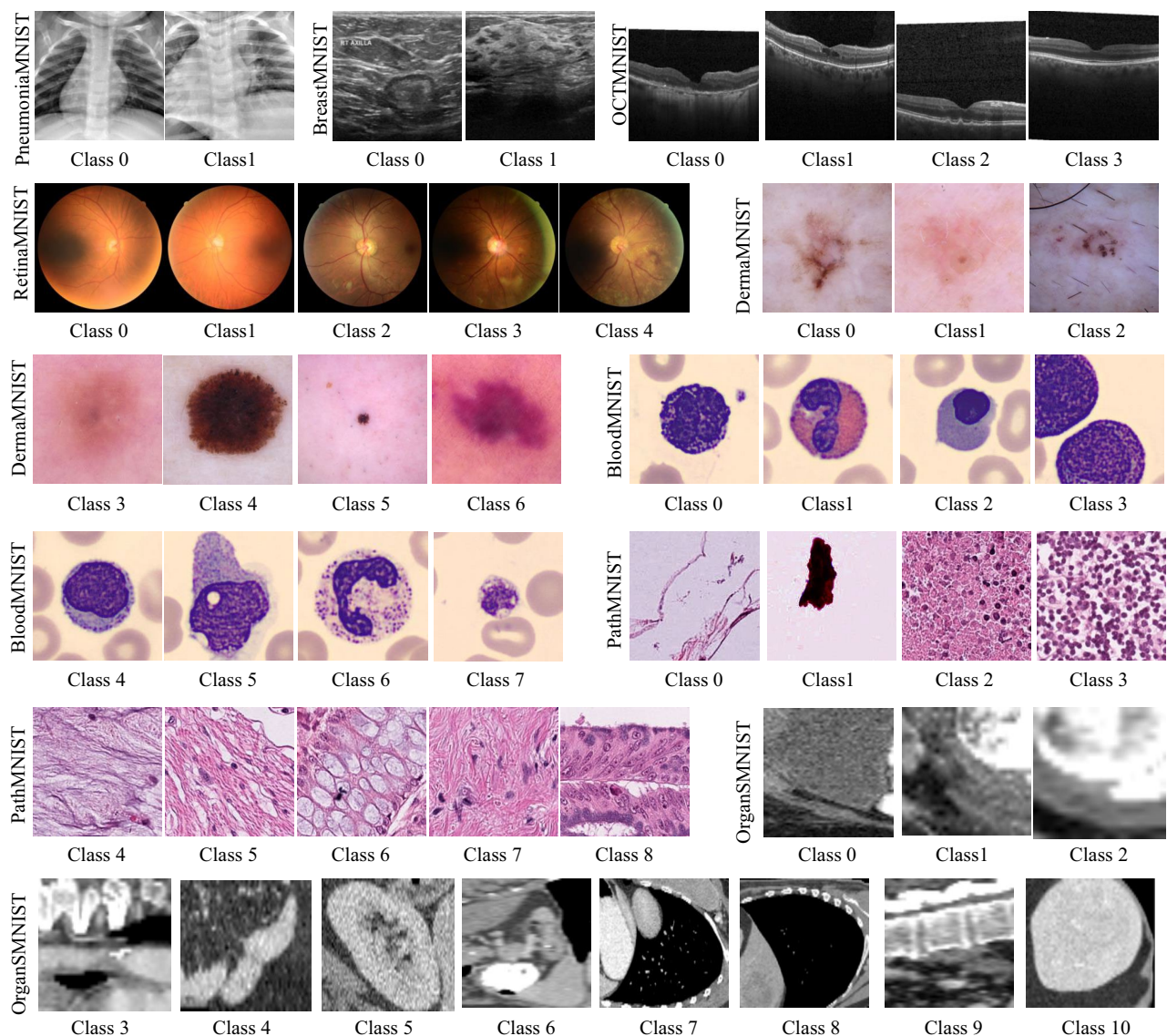


Fig. 1. MedMNIST datasets for 2D biomedical image classification.

ACC = Number of correct predictions/Total number of samples. To calculate the AUC metric for multi-class datasets, we first applied the softmax activation function to convert the raw expectation values into class probabilities. This ensures that the outputs for each test sample sum to 1 and can be interpreted as probabilities for each class. Then, we use the AUC function⁴⁰ from scikit-learn to calculate the AUC.

Noiseless results

We present the noiseless results of the QML framework using a simulator based on ACC and AUC metrics under different settings of input embedding features and quantum circuit trainable parameters for the MedMNIST datasets in Tables 2 and 3. The QML evaluation is performed by considering all the training and testing samples of the MedMNIST datasets as given in Table 1. The bold values in Tables 2 and 3 represent the best accuracy and AUC values. For the PneumoniaMNIST dataset, the QML model with 4 qubits, 49 input embedding features, and 120 trainable parameters provides the highest classification accuracy of 85.26% and AUC of 81.97% as compared to the other settings as shown in Table 2. In the case of the BreastMNIST dataset, we got the best accuracy of

Dataset	Number of qubits	Number of embedding gates	Number of trainable parameters	Noiseless ACC	Noiseless AUC
PneumoniaMNIST	4	49	60	0.8478	0.8098
			80	0.8462	0.8137
			100	0.8462	0.8068
			120	0.8526	0.8197
		64	60	0.8253	0.785
			80	0.8462	0.8128
			100	0.8526	0.812
			120	0.8478	0.8132
BreastMNIST	4	49	60	0.769	0.652
				0.801	0.684
			100	0.814	0.674
			120	0.808	0.693
		64	60	0.776	0.681
			80	0.769	0.722
			100	0.775	0.658
			120	0.794	0.694
OCTMNIST	4	49	60	0.433	0.657
			80	0.41	0.656
			100	0.404	0.648
			120	0.415	0.657
		64	60	0.388	0.628
			80	0.409	0.631
			100	0.423	0.651
			120	0.413	0.648
	8	49	60	0.387	0.62
			80	0.379	0.623
			100	0.443	0.659
			120	0.441	0.652
RetinaMNIST	5	49	60	0.4775	0.6268
			80	0.48	0.6646
			100	0.485	0.6434
			120	0.5	0.6458
		64	60	0.5025	0.6648
			80	0.505	0.6474
			100	0.5025	0.6586
			120	0.5175	0.6559

Table 2. Performance analysis of QML framework on simulator based on classification accuracy (ACC) and area under the curve (AUC) metrics under different configurations of quantum models. Bold values represent the best results.

Dataset	Number of qubits	Number of embedding gates	Number of trainable parameters	Noiseless ACC	Noiseless AUC
DermaMNIST	7	49	60	0.6688	0.6873
			80	0.6698	0.7144
			100	0.6698	0.7013
			120	0.6693	0.7059
		64	60	0.6723	0.6938
			80	0.6693	0.7094
			100	0.6728	0.6959
			120	0.6738	0.7343
BloodMNIST	8	49	60	0.415	0.774
			80	0.371	0.781
			100	0.462	0.811
			120	0.432	0.782
		64	60	0.44	0.792
			80	0.434	0.796
			100	0.421	0.791
			120	0.482	0.821
PathMNIST	9	49	60	0.3398	0.7542
			80	0.3553	0.7492
			100	0.3467	0.7703
			120	0.3614	0.7796
		64	60	0.3443	0.7574
			80	0.3341	0.7708
			100	0.3558	0.7682
			120	0.3723	0.784
OrganSMNIST	11	49	60	0.3735	0.7492
			80	0.3711	0.7585
			100	0.369	0.7624
			120	0.3544	0.7689
		64	60	0.3353	0.729
			80	0.3528	0.7709
			100	0.3786	0.7599
			120	0.3681	0.7802

Table 3. Performance analysis of QML framework on simulator based on classification accuracy (ACC) and area under the curve (AUC) metrics under different configurations of quantum models. Bold values represent the best results.

81.40% for the 4-qubit quantum model having 49 embedding features and 100 trainable parameters, while the highest AUC of 72.20% is achieved for the model with 64 embedding features and 80 trainable parameters. Similarly, we got the highest accuracy of 44.30% and AUC of 65.90% for an 8-qubit quantum model having 49 embedding features and 100 trainable parameters for the OCTMNIST dataset. For the RetinaMNIST dataset, we achieved the best classification accuracy of 51.75% for the 5-qubit quantum model having 64 input embedding features and 120 trainable parameters. The highest AUC of 66.48% is obtained for the quantum model having 64 embedding features and 60 trainable parameters as depicted in Table 2. A 7-qubit quantum model with 64 input embedding features and 120 trainable parameters provides the highest accuracy of 67.38% and AUC of 73.43% for the 7-class DermaMNIST dataset as shown in Table 3. For an 8-class BloodMNIST dataset, we achieved the highest accuracy of 48.20% and AUC of 82.10% using the QML model with 64 embedding features and 120 trainable parameters. Similarly, we got the best accuracy of 37.23% and AUC of 78.40% for a 9-class PathMNIST dataset using the QML model having 64 embedding features and 120 trainable parameters. Lastly, the QML model with 64 features and 100 trainable parameters provides the best classification accuracy of 37.86% for the 11-class OrganSMNIST dataset, while the best AUC of 78.02% is achieved for the QML model having 64 embedding features and 120 trainable parameters. It is clear from these results that the quantum models without any classical neural networks and with very limited input features show promising capability to classify medical imaging datasets. Also, it is observed from the Tables 2 and 3 that there is a significant difference between the accuracy and AUC values, for instance, we got an accuracy of 48.20% and AUC of 82.10% for the BloodMNIST dataset. A similar trend can be seen for the PathMNIST and OrganSMNIST datasets. Note that we considered AUC as our primary performance evaluation metric because it is less sensitive to the class imbalance of the MedMNIST datasets as compared to the accuracy metric. Moreover, it can be noted that for most of the multi-class datasets, we got the best results corresponding to the QML models having more embedding features and trainable parameters in the quantum circuit.

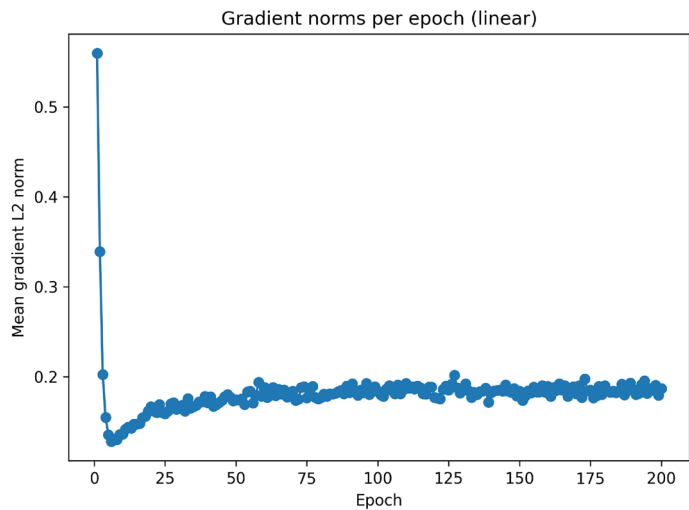


Fig. 2. Gradient L2 norm vs epoch plot during the training phase of 7-qubit circuit used for DermaMNIST dataset.

	IBM Cleveland (ACC)	IBM Cleveland (AUC)
No error suppression/mitigation	0.286	0.545
Error suppression (DD+Twirling)	0.33	0.591
M3 error mitigation	0.285	0.533
DD+Twirling+M3	0.397	0.622

Table 4. Ablation of different error suppression/mitigation techniques using OCTMNIST dataset on real IBM Cleveland hardware. Bold values represent the best results.

To demonstrate robustness and reliability of the noiseless results, we conducted three independent training trials of the same 7-qubit quantum model used for DermaMNIST dataset and observed 0.6707 ± 0.0032 accuracy and 0.6882 ± 0.0113 AUC, confirming that the optimization procedure is stable across runs. Note that we are not able to provide the repeated trials variance for all the considered MedMNIST datasets because training of the quantum models using large number of training samples is time-consuming.

Moreover, it is worth mentioning that variational quantum circuits are known to exhibit vanishing gradients as the number of qubits or circuit depth increases, a phenomenon commonly referred to as the barren plateau problem. In this study, we used quantum circuits with a shallow width and depth, locality-preserving connectivity, which substantially reduces the likelihood of entering such regimes. For instance, during the training of 7-qubit circuit used for DermaMNIST dataset, we monitored the L2 norm of the parameter gradients per epoch and observed non-zero, stable magnitudes throughout optimization as shown in Fig. 2. This empirical behavior indicates that, for the circuit size and dataset considered in this study, gradient vanishing was not a limiting factor. Nonetheless, for larger circuits the use of problem-informed ansatz or layerwise/iterative training would be advisable.

Ablation study of error suppression and mitigation

We also performed an ablation study on IBM Cleveland to understand the impact of error suppression and error mitigation techniques on the performance of QML models for medical imaging classification as shown in Table 4. We performed this ablation on the OCTMNIST dataset because of the significant effect of hardware noise on the performance of the QML model. For the OCTMNIST dataset, the QML workflow provides the noiseless accuracy of 44.30% and AUC of 65.90% as shown in Table 2. When we ran this quantum model on real IBM hardware without any error suppression and mitigation techniques, we got an ACC of 28.60% and an AUC of 54.50%. Therefore, we observed a reduction of 15.70% and 11.40% in accuracy and AUC, respectively due to the noise of IBM Cleveland hardware. In this ablation, since we are using a sampler in the inference stage, we considered two error suppression techniques i.e., DD and Twir and one error mitigation technique i.e., M3 mitigation. It is observed from the Table 4 that when we use both error suppression techniques and M3 mitigation together (DD+Twirling+M3), we got the best accuracy of 39.70% and AUC of 62.20%, thereby providing an improvement of 11.10% and 7.7% in accuracy and AUC respectively when compared to the case where no suppression and mitigation is used. Based on this ablation study on the OCTMNIST dataset, we used this combination (DD+Twirling+M3) for all the MedMNIST datasets we studied.

IBM Cleveland hardware results

We selected the best performing QML models from Tables 2 and 3 based on the AUC metric for inference on IBM Cleveland for all considered MedMNIST datasets. Due to hardware limitations, we considered 1000 testing samples to evaluate the QML models for the datasets having a large number of testing samples. We presented noiseless and IBM hardware results (with and without error suppression and mitigation) together in Table 5 along with the specifications of the quantum models such as number of qubits, number of embedded features/gates, trainable parameters, quantum circuit gates, and circuit depth. It is observed from Table 5 that even with the small number of input embedding features, the QML models consistently provide promising results in terms of classification accuracy and AUC metrics. It is worth noting that the noiseless results and hardware results (without error suppression and mitigation) are quite comparable for most of the MedMNIST datasets except the OCTMNIST and OrganSMNIST datasets. This is because we generated device-aware quantum circuits with optimal circuit gates, depth, and robustness against hardware noise. For the OCTMNIST dataset, the classification accuracy and AUC on the hardware (without error suppression and mitigation) are reduced by 15.70% and 11.40% respectively as compared to the noiseless results. Similarly, accuracy and AUC decreased by 14.11% and 4.8% respectively for the OrganSMNIST dataset as compared to the noiseless results. This indicates that there is scope for further improvement in the design of device-aware quantum circuits. To mitigate the effect of hardware noise, we ran the hardware experiments for the selected MedMNIST datasets using an error suppression and error mitigation combination (DD+Twirling+M3) based on the ablation study conducted in Table 4. We noted improvements in classification accuracy and AUC results using the DD+Twirling+M3 combination as compared to the hardware results without error suppression and mitigation for all the datasets except the BreastMNIST dataset. For instance, the classification accuracy and AUC values are significantly improved by 11.10% and 7.7% respectively as compared to the hardware results without error suppression and mitigation for the OCTMNIST dataset. For the BreastMNIST dataset, we observed that the accuracy and AUC values decreased by 1.3% and 2.4% respectively when compared to the hardware results without error suppression and mitigation. This phenomenon arises primarily due to the dense structure of the BreastMNIST quantum circuit. Specifically, the BreastMNIST circuit (using 4 qubits) is compact with a circuit depth of 59 (highest among all datasets, as observed in Table 5) and densely populated with 187 total gates. In such densely packed circuits, DD pulses which are effective mainly when qubits have extended idle periods may not help and instead can introduce additional noise through imperfect pulses. Our findings align well with the insights presented by IBM Quantum in their error mitigation guide⁴¹, which notes explicitly that DD can negatively impact dense circuits due to pulse-induced noise.

To assess the stability of the proposed QML model under realistic device noise, we executed our DermaMNIST experiment three times on the IBM Cleveland backend. The three hardware runs yielded 67.7%, 66.60%, and 66.90% accuracy, and AUCs of 69.8%, 72.89%, and 72.57%, respectively, indicating a tight clustering of results (mean accuracy $\approx$ 67.1% with a standard deviation of $\approx$ 0.6 percentage points; mean AUC $\approx$ 71.8% with a standard deviation of $\approx$ 1.7 percentage points). This shows that the hardware evaluation is numerically stable across repeated executions on the same backend, with only small variation that can plausibly be attributed to day-to-day noise-performance differences on the device. At the same time, we want to clarify that access to real quantum hardware is limited and costly and running many hardware trials for every dataset considered in this work is not feasible in our current access model.

Comparison with classical ML models

We also provided a comparative analysis with the benchmarks of the classical ML models provided in MedMNIST¹ as shown in Table 6. To the best of our knowledge, this benchmarking study is the first to provide a comprehensive evaluation of QML (without using any classical neural networks) on real IBM quantum hardware using large-scale MedMNIST datasets. Although the classification results obtained by our QML workflow do not outperform the best-performing classical ML model due to the inherent limitations of current NISQ hardware, our results represent a significant step forward in evaluating QML for realistic applications in medical imaging. The classical ML models leverage the full spatial dimensions of input images and millions of trainable parameters, whereas the QML model operates on a significantly reduced feature set using very limited

Dataset	Qubits	Embed gates	Tunable params	Circuit gates	Circuit depth	Noiseless		IBM Cleveland			
						ACC	AUC	No supp./miti.		DD+Twir+M3	
						ACC	AUC	ACC	AUC	ACC	AUC
PneumoniaMNIST	4	49	120	184	51	0.8526	0.8197	0.8381	0.8038	0.8542	0.8218
BreastMNIST	4	64	80	187	59	0.769	0.722	0.782	0.738	0.769	0.714
OCTMNIST	8	49	100	208	47	0.443	0.659	0.286	0.545	0.397	0.622
RetinaMNIST	5	64	60	140	29	0.5025	0.6648	0.5025	0.6638	0.505	0.6642
DermaMNIST	7	64	120	212	33	0.6738	0.7343	0.671	0.6825	0.677	0.6982
BloodMNIST	8	64	120	211	27	0.482	0.821	0.422	0.824	0.424	0.826
PathMNIST	9	64	120	218	27	0.3723	0.784	0.35	0.7544	0.418	0.7795
OrganSMNIST	11	64	120	213	23	0.3681	0.7802	0.227	0.7322	0.34	0.7678

Table 5. Performance analysis of QML framework on real IBM Cleveland hardware based on classification accuracy (ACC) and area under the curve (AUC) metrics for different MedMNIST datasets.

Method (Input size)	PneumoniaMNIST		BreastMNIST		OCTMNIST		RetinaMNIST	
	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
ResNet-18 (28×28×3) ⁴²	0.944	0.854	0.901	0.863	0.943	0.743	0.717	0.524
ResNet-18 (224×224×3) ⁴²	0.956	0.864	0.891	0.833	0.958	0.763	0.71	0.493
ResNet-50 (28×28×3) ⁴²	0.948	0.854	0.857	0.812	0.952	0.762	0.726	0.528
ResNet-50 (224×224×3) ⁴²	0.962	0.884	0.866	0.842	0.958	0.776	0.716	0.511
Auto-sklearn (28×28×3) ⁴³	0.942	0.855	0.836	0.803	0.887	0.601	0.69	0.515
AutoKeras (28×28×3) ⁴⁴	0.947	0.878	0.871	0.831	0.955	0.763	0.719	0.503
Google AutoML Vision (28×28×3) ⁴⁵	0.991	0.946	0.919	0.861	0.963	0.771	0.75	0.531
QML model (7×7 or 8×8) (IBM Cleveland)	0.822	0.854	0.714	0.769	0.622	0.397	0.664	0.505

Method (Input size)	DermaMNIST		BloodMNIST		PathMNIST		OrganSMNIST	
	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
ResNet-18 (28×28×3) ⁴²	0.917	0.735	0.998	0.958	0.983	0.907	0.972	0.782
ResNet-18 (224×224×3) ⁴²	0.92	0.754	0.998	0.963	0.989	0.909	0.974	0.778
ResNet-50 (28×28×3) ⁴²	0.913	0.735	0.997	0.956	0.99	0.911	0.972	0.77
ResNet-50 (224×224×3) ⁴²	0.912	0.731	0.997	0.95	0.989	0.892	0.975	0.785
Auto-sklearn (28×28×3) ⁴³	0.902	0.719	0.984	0.878	0.934	0.716	0.945	0.672
AutoKeras (28×28×3) ⁴⁴	0.915	0.749	0.998	0.961	0.959	0.834	0.974	0.813
Google AutoML Vision (28×28×3) ⁴⁵	0.914	0.768	0.998	0.966	0.944	0.728	0.964	0.749
QML model (7×7 or 8×8) (IBM Cleveland)	0.698	0.677	0.826	0.424	0.78	0.418	0.768	0.34

Table 6. Comparative analysis of QML model and classical benchmark ML models using AUC and ACC metrics on MedMNIST datasets. All the classical ML baselines use full resolution input images but our QML model uses downsampled 7×7 or 8×8 image features due to the current hardware limitations. Bold values represent the best results.

number of trainable parameters, dictated by quantum hardware constraints. Despite this, the QML model achieves promising results on IBM Cleveland in terms of classification accuracy and AUC values. For the 2-class PneumoniaMNIST dataset, we obtained good AUC and classification accuracies of 82.2% and 85.4% respectively. The QML accuracy of 85.4% for the PneumoniaMNIST dataset is equal to the accuracy obtained by the classical ML models including ResNet-18 (28)⁴² and ResNet-50 (28)⁴² as shown in Table 6. Also, the QML accuracy 85.4% for PneumoniaMNIST dataset is lagging by only 0.1%, 1%, 2.4%, and 3% when compared with the Auto-sklearn⁴³, ResNet-18 (224)⁴², AutoKeras⁴⁴, and ResNet-50 (224)⁴² classical ML models, respectively. Similarly, our QML model classification accuracy is lagging by 3.4% when compared to the Auto-sklearn⁴³ classical ML model for the BreastMNIST dataset. Moreover, in the case of the 5-class RetinaMNIST dataset, we achieved an AUC and accuracy of 66.4% and 50.5% respectively which lags by 8.6% and 2.6% in AUC and accuracy when compared to the best performing classical Google AutoML Vision⁴⁵ model. The AUC value 66.4% of the QML model for RetinaMNIST lags only by 2.6% compared to the Auto-sklearn⁴³ classical ML model. Importantly, the QML model accuracy of 50.5% for RetinaMNIST outperforms the ResNet-18 (224)⁴² and AutoKeras⁴⁴ classical ML models by 1.2% and 0.2%, respectively. Even for the 7-class DermaMNIST dataset, the QML model obtained good results with AUC and accuracy values of 69.8% and 67.7% respectively. The accuracy value achieved by the QML model for the DermaMNIST dataset only lags by 4.2% and 9.1% when compared to the Auto-sklearn⁴³ model and the best performing classical Google AutoML Vision⁴⁵ ML model, respectively. The QML model can achieve an AUC of more than 70% for all the MedMNIST datasets we explored except for the OCTMNIST, RetinaMNIST, and DermaMNIST datasets. It is worth mentioning that for the 2-class PneumoniaMNIST and 8-class BloodMNIST datasets, we obtained AUCs of more than 80%. These results underscore the potential of QML in image classification, particularly when benchmarked against the existing literature on image classification using quantum approaches, where achieving comparable performance with classical models remains an open challenge. Our study provides a critical evaluation of QML for realistic applications, showcasing its capability even in the current phase of quantum computing. By systematically benchmarking QML performance on IBM Cleveland across diverse MedMNIST datasets, this work paves the way for future advancements in QML and highlights its potential to complement classical ML in the years to come.

We further conduct a more feasible comparison to validate the performance of our QML model by training different lightweight classical ML classifiers including logistic regression, SVM, and shallow MLPs with different number of hidden layers on the same down-scaled DermaMNIST-7 features ($8 \times 8 = 64$) as used in our QML model as shown in Table 7. All the considered classical baselines and QML end up in a narrow performance band around 67–70% in accuracy (ACC). Our QML model matches or slightly improves on the decision tree, is close to logistic regression in ACC, and does so with only 120 trainable parameters. QML AUC (0.70) is below the strongest small MLPs or SVM, but still within the range that these simple classical methods achieve on the

Model	Trainable parameters	AUC	ACC
Logistic regression	–	0.7543	0.6693
Decision Tree	–	0.7068	0.6653
SVM	–	0.7910	0.6703
MLP (16)	1159	0.7738	0.6763
MLP (32)	2311	0.7810	0.6793
MLP (64)	4615	0.7946	0.6868
MLP (128)	9223	0.8021	0.6918
MLP (256)	18439	0.7959	0.6818
MLP (128, 64)	17031	0.8029	0.6988
MLP (256, 128)	50439	0.7978	0.6918
MLP (256, 128, 64)	58247	0.7828	0.6698
QML model (ibm_cleveland)	120	0.70	0.677

Table 7. Performance comparison of lightweight classical ML baselines and the proposed QML model on the same downscaled DermaMNIST-7 features (8×8).

same features. The highest classical scores at this resolution ($\text{AUC} \approx 0.80$, $\text{ACC} \approx 69\text{--}70\%$) come from MLPs with on the order of 9k–17k+ parameters, whereas the QML circuit uses only 120 parameters. This suggests that the quantum model has a favorable “accuracy-per-parameter” profile on low-information inputs: it can express a useful decision boundary with far fewer tunable degrees of freedom. These results indicate that the QML has the potential to achieve better representational capability as compared to the classical counterparts, but due to the current hardware limitations we cannot explore it in this study.

Discussion

This study highlights the potential of QML in medical imaging by benchmarking the MedMNIST datasets on real quantum hardware. We successfully trained and evaluated noise-resilient quantum circuits using advanced error suppression and mitigation techniques to address the challenges posed by hardware noise. The results demonstrate the feasibility of applying QML to complex, real-world datasets under current NISQ conditions. Despite the use of a limited number of input embedding features, the QML models achieved promising classification performance across binary and multi-class tasks. For example, on the 2-class PneumoniaMNIST dataset, the best QML model achieved an accuracy of 85.4% and an AUC of 82.2%. On the 5-class RetinaMNIST dataset, we obtained results comparable to classical baselines, with an accuracy of 50.5% and an AUC of 66.4%. For the 7-class DermaMNIST dataset, the QML model achieved 67.7% accuracy and 69.8% AUC, further demonstrating its capability for multi-class classification. These findings underscore the growing maturity of quantum computing for practical applications and highlight the importance of continued research into scalable quantum models and improved circuit design. It is also important to note that although the quantum circuits used in this study are classically simulable, the proposed QML framework establishes a critical foundation for future research with large-scale, classically intractable quantum circuits paving the way toward practical quantum advantage in real-world applications.

Despite encouraging results, this study has some limitations. The quantum circuits employed are relatively small using 4 to 11 qubits, 140 to 218 gates, and depths of 23 to 59, which limits their ability to fully exploit quantum computational advantages. Additionally, due to current hardware constraints such as noise, limited coherence time, and restricted access to large-scale systems, we capped the number of embedded input features at 64, reducing the expressive power of the models. Future work should focus on scaling up circuit size, depth, and feature embedding as quantum hardware matures, enabling richer data representations and revealing potential quantum advantages. Further, advancing circuit design through application-specific embedding strategies, expressive ansatz, and hardware-aware error mitigation can enhance QML performance and reliability in real-world medical imaging tasks.

Methods

In this section, we discuss our QML methodology for medical image classification, as illustrated in Fig. 3. Our approach is structured into four key stages: data preprocessing, generation of device-aware quantum circuits, training of the quantum circuits on classical hardware using a training dataset, and inference of the trained quantum model on real IBM quantum hardware using a testing dataset. It is important to emphasize that, due to the limitations of current NISQ quantum devices, training quantum models with large datasets directly on hardware remains infeasible. To address this challenge, the QML workflow adopts a hybrid strategy, where the training phase is executed on classical hardware, leveraging their computational power, while the inference stage is performed on real quantum hardware to validate the model’s performance. The following subsections provide a detailed explanation of each stage of the QML methodology.

Data preprocessing

The preprocessing stage focuses on reducing the feature dimensionality of input images, a necessary step in the QML workflow due to the constraints of current quantum hardware. Specifically, we employ angle encoding

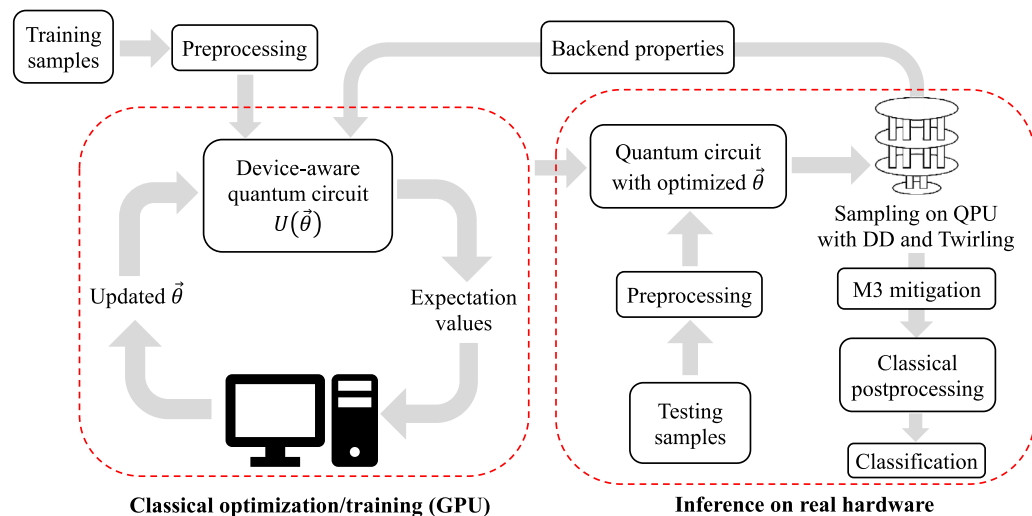


Fig. 3. QML framework for medical image classification.

to map classical data to quantum states. The angle encoding method is limited by the hardware's inability to handle high-dimensional feature spaces in the current NISQ era. Therefore, it is very important to reduce the spatial dimension of the images effectively. We used average pooling to downscale the input image dimension from 224×224 to 7×7 and 8×8 for the considered MedMNIST datasets, thereby providing 49 and 64 features, respectively. This indicates that performing angle encoding for 49 and 64 features requires the incorporation of 49 and 64 Pauli rotation gates, respectively, within the quantum circuit. The quantum circuit representational capacity should be adjusted accordingly to effectively capture the patterns in the input features. Post-feature reduction, the pixel values are normalized to the range $[0, \pi]$, aligning with the requirements of angle encoding for efficient quantum state preparation. It is important to note that this preprocessing stage introduces information loss compared to classical machine learning models, which can process entire images as input without dimensionality reduction. In a QML workflow, this limitation arises due to the constraints of current quantum hardware, which restrict the number of features that can be encoded into quantum states. Consequently, while classical models can leverage the full spatial resolution of input images, quantum models operate with a reduced feature set, potentially impacting the overall representational capacity. This trade-off underscores the need for effective feature-reduction techniques that preserve essential information while ensuring compatibility with the limitations of quantum devices.

Efficient and noise-resilient quantum circuits for QML

The second crucial step of the QML framework involves designing a noise-robust and efficient quantum circuit that is expressively powerful enough to learn meaningful patterns from the input data for the medical imaging classification tasks. In the current NISQ era, developing device-aware quantum circuits tailored to the target backend properties, such as qubit connectivity and gate fidelities, has emerged as a key strategy to mitigate the impact of hardware noise. However, designing such circuits remains a significant challenge due to the lack of well-defined guidelines for quantum circuit architecture in QML workflows. Additionally, the optimal design of a quantum circuit is highly dataset-dependent, as varying datasets require different levels of expressivity to effectively capture their underlying patterns.

To address these challenges, we incorporated the Élivágar framework⁴⁶ into the QML workflow to generate hardware-aware quantum circuits tailored to our medical image classification task. This framework allowed us to create noise-resilient and computationally efficient quantum circuits that align with the physical constraints of the target quantum hardware while maximizing model performance. The first step of the Élivágar framework is to generate candidate quantum circuits using information from the target quantum device. Élivágar samples candidate circuits directly on connected subgraphs of the target device and every candidate circuit was grown to respect the actual 1- and 2-qubit couplings of target quantum device. This hardware-aware growth minimizes the need for additional SWAP operations, keeps the two-qubit depth in a feasible range, and ultimately helps narrow the gap between noiseless simulation and on-hardware runs. Moreover, the local entanglement between hardware-adjacent qubits helps the quantum circuit to model correlations between nearby encoded pixels without incurring the noise penalty of long-range entanglement. We generated 250 quantum circuits in all the MedMNIST dataset experiments. These quantum circuits include three key components: (1) data encoding gates, (2) a variational or ansatz circuit, and (3) measurement operations that map quantum states to classical bits. The data encoding layer leverages parameterized rotation gates to embed classical input data into quantum states. The ansatz, or variational circuit, contains trainable Pauli rotation gates interspersed with CNOT gates to ensure qubit entanglement. The parameters of these gates (θ) are optimized during training to minimize the loss function.

In the second step, Elivágar evaluates candidate circuits in two stages that are aligned with our medical-imaging use case. First, it filters circuits using a Clifford Noise Resilience (CNR) score to discard topologies that would be overly sensitive to the noise profile of the device. Second, among the noise-robust candidates, it ranks circuits by their representational capacity (RepCap), i.e., their ability to produce quantum states that are separable across classes while preserving intra-class similarity. This is particularly important for MedMNIST datasets, which are multi-class and, in some cases imbalanced: we require a circuit that actually distinguishes classes rather than a generic, very expressive but noise-vulnerable ansatz. CNR⁴⁶ computation process involves the generation and execution of multiple Clifford replicas of the circuit (C) denoted by $C_R(i)$, followed by a fidelity assessment between the noisy (P_{noisy}) and noiseless ($P_{noiseless}$) outputs. To achieve this, the Total Variation Distance (TVD) is first evaluated between noiseless and noise outputs as defined in the Eq. (1) and the circuit fidelity (F) is computed by $F = 1 - TVD$.

$$TVD = \frac{1}{2} \sum_{i \in [2^{n_q}]} |P_{noiseless}(|i\rangle) - P_{noisy}(|i\rangle)| \quad (1)$$

where n_q denotes the number of qubits in the circuit. The CNR of a circuit is then computed as the average fidelity of the M Clifford replicas defined as:

$$CNR(C) = \frac{1}{M} \sum_{i=1}^M F(P_{noiseless}^{C_R(i)}, P_{noisy}^{C_R(i)}) \quad (2)$$

The low-fidelity candidate circuits with CNR values lower than the threshold value (0.7) are excluded from further evaluation. While this process is typically conducted on a quantum device, it is time-intensive due to the need to run thousands of circuits, often facing significant queuing delays. To streamline the process, we performed CNR calculations using IBM Cleveland's hardware noise model on a simulator (with the number of shots = 10000 and $M = 32$), enabling efficient evaluation of circuit noise resilience while saving considerable time. Afterward, the performance of selected quantum circuits is assessed using RepCap⁴⁶ defined as:

$$RepCap(C) = 1 - \frac{\|R_C - R_{ref}\|_2^2}{2n_c d_c^2} \quad (3)$$

where d_c represents the number of samples selected per class, n_c is the number of classes in the training dataset, and $\|\cdot\|_2$ denotes the Frobenius norm. We used $d_c = 16$ for all the MedMNIST dataset experiments. The matrix R_C encodes the pairwise similarities between the quantum representations of data samples, while R_{ref} is a reference matrix capturing the ideal behavior of the circuit, where $R_{ref}(i, j) = 1$ if two samples belong to the same class and 0 otherwise. A higher RepCap value suggests that the circuit has a greater capacity to distinguish between different classes in the dataset. Finally, a composite scoring metric (F_{score}) is computed based on CNR and RepCap to effectively select the most suitable quantum circuit defined as:

$$F_{score}(C) = CNR(C)^{\alpha_{CNR}} \times RepCap(C) \quad (4)$$

where, α_{CNR} is a tunable hyperparameter that controls the relative weight of noise resilience in the scoring process and we set $\alpha_{CNR} = 0.5$ in our MedMNIST experiments.

To ensure adaptability and effectiveness, we designed multiple quantum circuits with varying levels of expressivity by systematically adjusting the number of trainable parameters and the depth of the ansatz. These circuits were carefully tailored to meet the unique requirements of the MedMNIST datasets. Specifically, we generated quantum circuits with different configurations, incorporating varying numbers of trainable parameters such as 60, 80, 100, and 120 for all the MedMNIST datasets to ensure fairness and reproducibility. Our parameter choices were inspired by the Elivágar framework⁴⁶, which explored 16–72 parameters on simpler MNIST tasks. Given the higher complexity of MedMNIST, we extended this range based on preliminary experiments to accommodate both binary and multi-class settings. We further performed an ablation study (see Tables 2 and 3) to analyze how model expressiveness, as reflected in parameter count, correlates with classification performance. Notably, we capped the maximum parameter count at 120 to stay within the computational and noise-tolerance limits of current NISQ devices. As an example, we provided a complete layout for the quantum circuit generated by the Elivágar framework for the PneumoniaMNIST dataset, which achieved the best performance in the noiseless (ideal) simulation setting, as shown in Fig. 4 to provide more clarity and enhanced reproducibility. It can be observed that this quantum circuit has a gate composition of one-qubit rotation gates (R_z , R_y , and R_x), two-qubit ECR gates and some additional gates including X and SX gates. The number of R_z , R_y , R_x , ECR, X , and SX gates are 59, 57, 53, 7, 4, and 3 respectively. The depth of the circuit and ECR gates are 51 and 7 respectively. The quantum circuits use Pauli-Z measurements on all qubits to extract expectation values. Thus, the observable used in each experiment is the tensor product of Pauli-Z operators acting on all qubits, and the resulting vector of expectation values is utilized directly for downstream classification tasks. Importantly, to thoroughly investigate the potential of quantum circuits in medical imaging classification, the QML framework exclusively employs quantum models without incorporating any classical neural network components. This approach highlights the intrinsic capabilities of quantum circuits and paves the way for future advancements in leveraging QML for medical imaging tasks.



Fig. 4. Quantum circuit generated by the Ėlivágar framework for PneumoniaMNIST dataset.

Training quantum model on classical hardware

The training phase of the QML framework for medical image classification is executed on classical hardware as illustrated in Fig. 3. This approach circumvents the current limitations of quantum hardware in the NISQ era, such as limited qubits, high noise levels, and decoherence effects by performing the computationally expensive training process on classical GPUs even for large-scale datasets. The QML workflow integrates classical optimization techniques with noise-resilient, hardware-efficient quantum circuits tailored for medical imaging tasks. The training pipeline includes iterative optimization using a gradient-based method and loss evaluation. The variational quantum circuit parameters (θ) are iteratively updated within a classical optimization loop to minimize a task-specific loss function. This iterative process continues until convergence is achieved or the maximum number of training epochs is reached, ensuring the parameters are optimized for the target classification task.

We conducted the training on noiseless simulators using the TorchQuantum library⁴⁷, which provides an ideal environment for parameter optimization without interference from hardware-induced noise. This approach significantly reduces computational overhead compared to direct hardware training and ensures consistent gradient computation. During each training iteration, the preprocessed input data is mapped to quantum states via encoding gates in the quantum circuit, which is then executed on the simulator. Expectation values obtained from circuit measurements are used to compute the loss function. Gradients of the loss function are calculated using backpropagation. Afterward, these gradients guide precise updates to the trainable parameters (θ). In our implementation, all the quantum circuits were trained for 200 epochs using the Adam optimizer with a learning rate of 0.01 and a batch size of 128, ensuring convergence to optimal parameter configurations. The MedMNIST contains datasets having a different number of classes. During training, we measure one qubit and two qubits for two class and four class datasets, respectively and a mean square loss function is used to calculate the loss. For the remaining multi-class datasets, we measure the number of qubits equal to the number of classes in the dataset, and the cross-entropy loss function is used to calculate the loss. To ensure compatibility with MedMNIST datasets, we trained circuits with varying numbers of trainable parameters, such as 60, 80, 100, and 120 for each dataset of MedMNIST to account for differences in dataset complexity. The best-performing models are saved along with their parameters, losses, and accuracies for later inference on real quantum hardware. We also performed inference of these trained quantum models on a simulator for all considered MedMNIST datasets. All quantum models are trained on an A100 Nvidia GPU with 43 GB RAM. Note that our objective in this study is to enable a fair comparison to the MedMNIST classical ML baselines. Therefore, we adopted the same train/test split protocol as the original benchmark and did not introduce additional cross-validation steps that were absent in the baseline. We acknowledge that repeated cross-validation would further strengthen statistical reliability, and we plan to investigate this in future work.

Inference on real IBM Cleveland hardware

The trained quantum models demonstrating the best classification performance on the simulator were selected for validation on the IBM Cleveland quantum hardware. During the inference stage of the QML framework, the trained quantum models with optimized parameters were utilized for evaluating the test samples. For each testing sample, the corresponding quantum circuit is dynamically constructed and transpiled at optimization level 3 to minimize circuit depth and gate count while ensuring compatibility with the hardware topology. Moreover, qubit connectivity was carefully configured to align with the hardware's topology, minimizing the need for SWAP gates, which can introduce unnecessary noise and overhead. To further enhance the quantum model performance, we employed error suppression techniques, i.e. DD and Twir during the sampling process on real IBM hardware. The DD is used to suppress the decoherence effects introduced due to the unwanted interaction between the qubits and the environment. Twir applies random single-qubit gates around a target gate to average out coherent errors. The sampling on real hardware is performed using Qiskit SamplerV2 in Session execution mode. Moreover, we also integrated M3 error mitigation⁴⁸ in our QML workflow. M3 mitigation is a scalable quantum error correction approach that operates within the noisy bitstring subspace, correcting correlated and uncorrelated errors using a memory-efficient, matrix-free iterative solver. The integration of these error suppression and mitigation strategies enables quantum models to achieve optimal inference performance on real quantum hardware. Lastly, classical post-processing is applied on the mitigated bitstrings to obtain the predicted classification labels. To achieve this, the mitigated bitstrings are processed to compute qubit probabilities defined as:

$$P(q, m) = \frac{N_{q,m}}{N_{\text{shots}}}, \quad m \in \{0, 1\} \quad (5)$$

where $P(q, m)$ is the probability of measuring state m on qubit q , $N_{q,m}$ is the number of times the q -th qubit is measured in state $|m\rangle$ during the experiment. The measurement outcome $m = 0$ corresponds to the state $|0\rangle$ and $m = 1$ corresponds to the state $|1\rangle$. N_{shots} represents the total number of measurement shots and we used $N_{\text{shots}} = 32000$ for all the MedMNIST datasets. These probabilities provide the final expectation values for classification computed as:

$$\langle Z_q \rangle = P(q, 0) - P(q, 1) \quad (6)$$

where $\langle Z_q \rangle$ denotes the expectation value of the Pauli-Z operator for the q -th qubit, calculated as the difference between the probabilities of measuring 0 and 1. This results in a vector of real-valued expectation scores. For the multi-class datasets, the length of this vector is equal to the number of classes. The predicted class label is determined by selecting the index of the maximum component of this expectation value vector using $\hat{y} = \arg \max_i \langle Z_i \rangle$. This argmax strategy is aligned with the training objective: during circuit training, the expectation value corresponding to the correct class is encouraged to approach +1, while all others are pushed toward -1. By conducting inference on real quantum device, we demonstrated the feasibility of deploying noise-resilient and device-aware quantum circuits for practical medical image classification tasks.

Data availability

The MedMNIST datasets used in this benchmarking study can be accessed using <https://medmnist.com/>.

Code availability

All the related codes are available at https://github.com/gurinder-hub/QML_MedMNIST.

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Author contributions

G. Singh and H. Jin proposed and designed the study of benchmarking MedMNIST datasets using quantum machine learning on real IBM Cleveland hardware. G. Singh and H. Jin performed the experiments under the supervision of K.M. Merz. G. Singh wrote the manuscript, with inputs and contributions from all authors.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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